

$$= \frac{1 - (-2/3) + 1/3}{1 + (-2/3) + 1/3} = \frac{2}{2/3} = 3$$

∴ Required equation is

$$x^2 - 5x + P = 0 \text{ or } x^2 - 2x + 3 = 0$$

25. (a) Given, $px^2 + qx + r = 0$

Let the roots be α and β .

By given condition, $\beta = 2\alpha$

$$\text{Product of roots } (\alpha\beta) = \frac{r}{p} = 2\alpha^2$$

$$\Rightarrow \alpha^2 = \frac{r}{2p} \quad \dots(i)$$

$$\text{Sum of roots } (\alpha + \beta) = -\frac{q}{p} = 3\alpha \quad \dots(ii)$$

On squaring Eq. (ii), we get $9\alpha^2 = \frac{q^2}{p^2}$

$$\Rightarrow 9 \left(\frac{r}{2p} \right) = \frac{q^2}{p^2} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow 9rp = 2q^2$$

26. (a) Let roots be α, β , then $\frac{\alpha}{\beta} = \frac{m}{n}$
[given]

$$\alpha + \beta = -1, \alpha\beta = 1$$

$$\sqrt{\frac{m}{n}} + \sqrt{\frac{n}{m}} + 1 = \sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + 1$$

$$= \frac{\alpha + \beta}{\sqrt{\alpha\beta}} + 1 = \frac{-1}{\sqrt{1}} + 1 = -1 + 1 = 0$$

27. (c) Let $\sqrt{\frac{x}{1-x}} = y \Rightarrow \sqrt{\frac{1-x}{x}} = \frac{1}{y}$
∴ $y + \frac{1}{y} = \frac{13}{6} \Rightarrow (y^2 + 1)6 = 13y$

$$\Rightarrow 6y^2 - 13y + 6 = 0$$

$$\Rightarrow 6y^2 - 9y - 4y + 6 = 0$$

$$\Rightarrow 3y(2y - 3) - 2(2y - 3) = 0$$

$$\Rightarrow (3y - 2)(2y - 3) = 0$$

$$\therefore y = \frac{2}{3} \text{ and } \frac{3}{2}$$

$$\text{When, } y = \frac{2}{3} \Rightarrow \frac{x}{1-x} = \frac{4}{9}$$

$$\Rightarrow 9x = 4 - 4x \Rightarrow x = \frac{4}{13}$$

$$\text{When, } y = \frac{3}{2} \Rightarrow \frac{x}{1-x} = \frac{9}{4}$$

$$\Rightarrow 4x = 9 - 9x \Rightarrow x = \frac{9}{13}$$

28. (d) Given, $\frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x}$

$$\therefore \frac{1}{a+b+x} - \frac{1}{x} = \frac{1}{a} + \frac{1}{b}$$

$$\Rightarrow \frac{-(a+b)}{(a+b+x)x} = \frac{(a+b)}{ab}$$

$$\Rightarrow x^2 + (a+b)x + ab = 0$$

$$\Rightarrow (x+a)(x+b) = 0$$

$$\Rightarrow x = -a, -b$$

29. (d) $8 \sec^2 \phi - 6 \sec \phi + 1 = 0$

$$\Rightarrow 8 \sec^2 \phi - 4 \sec \phi - 2 \sec \phi + 1 = 0$$

$$\Rightarrow (4 \sec \phi - 1)(2 \sec \phi - 1) = 0$$

$$\Rightarrow \sec \phi = 1/4 \text{ or } \sec \phi = 1/2$$

$$\text{But } \sec \phi \geq 1 \text{ or } \sec \phi \leq -1$$

Hence, the equation has no solution.

30. (c) Here, $3^{2x^2 - 7x + 7} = 9 = 3^2$

On comparing, we get

$$2x^2 - 7x + 7 = 2$$

$$2x^2 - 7x + 5 = 0$$

$$\text{Here, } D = b^2 - 4ac = 49 - 4(2)(5) = 9$$

Since, $D > 0$, so it has two real roots.

31. (c) Here, $x = \sqrt{2} + 2 \Rightarrow x - 2 = \sqrt{2}$

On squaring both sides, we get

$$(x - 2)^2 = 2 \Rightarrow x^2 - 4x + 4 = 2$$

$$\Rightarrow x^2 - 4x + 2 = 0$$

32. (b) $\frac{x^2 - bx}{ax - c} = \frac{k - 1}{k + 1}$

$$\Rightarrow (x^2 - bx)(k + 1) = (k - 1)(ax - c)$$

$$\Rightarrow x^2k + x^2 - bxc - bx = kax - kc - ax + c$$

$$\Rightarrow (k + 1)x^2 - x(bk + b + ka - a) + kc - c = 0$$

$$\Rightarrow (k + 1)x^2 - x(bk + b + ka - a) + kc - c = 0$$

Since, roots are reciprocal to each other.

So, product = 1

$$\text{i.e. } \frac{kc - c}{(k + 1)} = 1 \Rightarrow kc - c = k + 1$$

$$\Rightarrow k(c - 1) = c + 1 \Rightarrow k = \frac{c + 1}{c - 1}$$

33. (b) Here, $\alpha + \beta = b/a$ and $\alpha\beta = b/a$

$$\text{So, } \sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} = \frac{\alpha + \beta}{\sqrt{\alpha\beta}} = \frac{b/a}{\sqrt{b/a}} = \sqrt{\frac{b}{a}}$$

34. (a) Given, $2x^2 - 7x + 3 = 0$

$$\therefore 2x^2 - 6x - x + 3 = 0$$

$$\Rightarrow 2x(x - 3) - 1(x - 3) = 0$$

$$\Rightarrow (2x - 1)(x - 3) = 0$$

$$\text{When, } x = \frac{1}{2}$$

$$4 \left(\frac{1}{2} \right)^2 + a \left(\frac{1}{2} \right) - 3 = 0$$

$$\Rightarrow 1 + \frac{a}{2} - 3 = 0 \Rightarrow \frac{a}{2} = 2 \Rightarrow a = 4$$

$$\text{When, } x = 3, 4(3)^2 + a(3) - 3 = 0$$

$$\Rightarrow 36 + 3a - 3 = 0 \Rightarrow a = -11$$

$$\therefore a = -11 \text{ or } 4$$

35. (c) Here, for equation $x^2 + bx + c = 0$

$$\alpha + \beta = -b \text{ and } \alpha\beta = c$$

$$\text{and for } x^2 + px + q = 0$$

$$k\alpha + k\beta = -p \text{ and } k^2\alpha\beta = q$$

$$\text{Now, } qb^2 = k^2\alpha\beta(\alpha + \beta)^2$$

$$= k^2(\alpha + \beta)^2\alpha\beta$$

$$= [k(\alpha + \beta)]^2\alpha\beta$$

$$\therefore qb^2 = p^2c$$

36. (b) Since, -4 is a root of

$$x^2 + px - 4 = 0$$

$$(-4)^2 + p(-4) - 4 = 0$$

$$\Rightarrow 16 - 4p - 4 = 0 \Rightarrow p = 3$$

As roots of $x^2 + px + q = 0$ are equal.

$$\therefore D = B^2 - 4AC = 0$$

$$\Rightarrow p^2 - 4q = 0 \text{ or } 3^2 - 4q = 0 \Rightarrow q = \frac{9}{4}$$

$$\text{So, } p = 3, q = \frac{9}{4}$$

37. (c) Here, $\alpha + \beta = 3$ and $\alpha\beta = 3/2$

$$\Rightarrow \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha} \right) + 3 \left(\frac{1}{\alpha} + \frac{1}{\beta} \right) + 2\alpha\beta$$

$$= \frac{\alpha^2 + \beta^2}{\alpha\beta} + \frac{3(\alpha + \beta)}{\alpha\beta} + 2\alpha\beta$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} + \frac{3(\alpha + \beta)}{\alpha\beta} + 2\alpha\beta$$

$$= \frac{3^2 - 2 \cdot 3/2}{3/2} + \frac{3 \cdot 3}{3/2} + 2 \left(\frac{3}{2} \right)$$

$$= \frac{6}{3/2} + \frac{9}{3/2} + 3 = 4 + 6 + 3 = 13$$

38. (b) Let α, β be the roots of

$$x^2 + 2ax + b = 0 \quad \dots(i)$$

$$\Rightarrow \alpha + \beta = -2a \text{ and } \alpha\beta = b.$$

By hypothesis, $|\alpha - \beta| \leq 2m$

$$\Rightarrow (\alpha - \beta)^2 \leq 4m^2$$

$$\Rightarrow (\alpha + \beta)^2 - 4\alpha\beta \leq 4m^2$$

$$\Rightarrow 4a^2 - 4b \leq 4m^2$$

$$\Rightarrow a^2 - b \leq m^2 \quad \dots(ii)$$

and discriminant of Eq. (i) is greater than 0.

$$\Rightarrow 4a^2 - 4b > 0 \Rightarrow b < a^2 \quad \dots(iii)$$

From Eqs. (ii) and (iii),

$$b \in [a^2 - m^2, a^2)$$

39. (b) Refer to example 15. [Rule 5]

40. (b) Refer to example 14. [Rule 4]

41. (b) Since, $\sin \theta$ and $\cos \theta$ are the roots of the equation $ax^2 - bx + c = 0$.

$$\therefore \sin \theta + \cos \theta = \frac{b}{a} \quad \dots(i)$$

$$\text{and } \sin \theta \cos \theta = \frac{c}{a}$$

On squaring both sides of Eq. (i), we get

$$\Rightarrow \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = \frac{b^2}{a^2}$$

$$\Rightarrow 1 + 2 \left(\frac{c}{a} \right) = \frac{b^2}{a^2} \Rightarrow 2 \left(\frac{c}{a} \right) = \frac{b^2 - a^2}{a^2}$$

$$\Rightarrow 2ac = b^2 - a^2 \Rightarrow a^2 - b^2 + 2ac = 0$$

- 42.** (d) The equation will have equal roots, if $B^2 - 4AC = 0$

$$\therefore (2nc)^2 - 4(1+n^2)(c^2 - a^2) = 0$$

$$\Rightarrow 4n^2 c^2 - 4(c^2 + n^2 c^2 - a^2 - n^2 a^2) = 0$$

$$\Rightarrow -4c^2 + 4a^2 + 4n^2 a^2 = 0$$

$$\Rightarrow c^2 = a^2(1 + n^2)$$

43. (d) a and c have the same sign opposite to that of b .

44. (d) Let the roots of equation $x^2 - 2x + 4 = 0$ be α and β .
Then, $\alpha + \beta = 2$ and $\alpha\beta = 4$
On taking, $\alpha \rightarrow 2\alpha$ and $\beta \rightarrow 2\beta$
 $2\alpha + 2\beta = 4$ and $2\alpha \cdot 2\beta = 4 \times 4 = 16$
So, the new equation becomes $x^2 - 4x + 16 = 0$

45. (a) Since, α and β are the roots of the equation $x^2 + px + q = 0$
 $\therefore \alpha + \beta = -p$ and $\alpha\beta = q$
Now, $-\alpha^{-1} - \beta^{-1} = -\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$
 $= -\left(\frac{\alpha + \beta}{\alpha\beta}\right) = \frac{p}{q}$
and $\left(-\frac{1}{\alpha}\right)\left(-\frac{1}{\beta}\right) = \frac{1}{\alpha\beta} = \frac{1}{q}$
Hence, required equation is $x^2 - (-\alpha^{-1} - \beta^{-1})x + (-\alpha\beta)^{-1} = 0$
 $(-\beta^{-1}) = 0$
 $\Rightarrow x^2 - \frac{p}{q}x + \frac{1}{q} = 0$
 $\Rightarrow qx^2 - px + 1 = 0$

46. (c) Since, one root of the equation $ax^2 + x - 3 = 0$ is -1 .
 $\therefore a(-1)^2 + (-1) - 3 = 0 \Rightarrow a = 4$
 $\therefore 4x^2 + x - 3 = 0$
Let other root of this equation be α .
Now, product of roots = $\frac{c}{a}$
 $\therefore -1 \cdot \alpha = -\frac{3}{4} \Rightarrow \alpha = \frac{3}{4}$

47. (b) Since, the roots of equation $(a^2 + b^2)x^2 - 2(ac + bd)x + (c^2 + d^2) = 0$ are equal.
 $\therefore B^2 = 4AC$
 $\Rightarrow 4(ac + bd)^2 = 4(a^2 + b^2)(c^2 + d^2)$
 $\Rightarrow a^2 c^2 + b^2 d^2 + 2abcd = a^2 c^2 + a^2 d^2 + b^2 c^2 + b^2 d^2$
 $\Rightarrow (ad - bc)^2 = 0$
 $\therefore ad = bc$

48. (a) Given, $\sqrt{\frac{x}{x+3}} - \sqrt{\frac{x+3}{x}} = -\frac{3}{2}$
Let, $y = \sqrt{\frac{x}{x+3}}$, then $\sqrt{\frac{x+3}{x}} = \frac{1}{y}$
 $y - \frac{1}{y} = -\frac{3}{2} \Rightarrow 2y^2 - 2 = -3y$
 $\Rightarrow 2y^2 + 3y - 2 = 0$
 $\Rightarrow 2y(y+2) - 1(y+2) = 0$
 $\Rightarrow (2y-1)(y+2) = 0$
If, $(2y-1) = 0$
 $\Rightarrow y = \frac{1}{2} \Rightarrow y = \sqrt{\frac{x}{x+3}} \Rightarrow \sqrt{\frac{x}{x+3}} = \frac{1}{2}$
On squaring both sides, we get $\frac{x}{x+3} = \frac{1}{4}$
 $\Rightarrow 4x = x+3 \Rightarrow x = 1$
or $y+2 = 0 \Rightarrow y = -2$
Since, y cannot be negative.
Hence, $x = 1$ is the required solution.

49. (c) Given, $4^x - 3 \cdot 2^{x+2} + 32 = 0$
 $\Rightarrow 2^{2x} - 3 \cdot 2^x \cdot 2^2 + 32 = 0$
 $\Rightarrow 2^{2x} - 12 \cdot 2^x + 32 = 0$
 $\Rightarrow 2^{2x} - 8 \cdot 2^x - 4 \cdot 2^x + 32 = 0$
 $\Rightarrow (2^x - 8)(2^x - 4) = 0$
Either, $2^x = 8 \Rightarrow x = 3$
or $2^x = 4 \Rightarrow x = 2$

50. (a) Since, α and β are the roots of the equation $x^2 - x - 1 = 0$.
 $\therefore \alpha + \beta = 1$ and $\alpha\beta = -1$
Now, $\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2$
 $= [(\alpha + \beta)^2 - 2\alpha\beta]^2 - 2(\alpha\beta)^2$
 $= (1+2)^2 - 2 = 9 - 2 = 7$

51. (c) Let the roots of equation $ax^2 + bx + c = 0$ be $-\alpha$ and $-\frac{1}{\alpha}$.
 $\therefore$ Product of roots = $\frac{c}{a}$
 $(-\alpha)\left(-\frac{1}{\alpha}\right) = \frac{c}{a} \Rightarrow 1 = \frac{c}{a} \Rightarrow c = a$

52. (d) Let the roots of the quadratic equation be α and β .
Then, $\alpha + \beta = 9$ and $\alpha\beta = 9$
Hence, equation is $x^2 - (\alpha + \beta)x + (\alpha\beta) = 0$
 $\Rightarrow x^2 - 9x + 9 = 0$

53. (c) Given, $\log_{10}(x^2 - 6x + 45) = 2$
 $\Rightarrow (x^2 - 6x + 45) = 10^2 = 100$
 $\Rightarrow x^2 - 6x - 55 = 0$
 $\Rightarrow x^2 - 11x + 5x - 55 = 0$
 $\Rightarrow x(x-11) + 5(x-11) = 0$
 $\Rightarrow (x+5)(x-11) = 0$
 $\therefore x = 11, -5$

54. (c) Given, $\frac{1}{x+a} + \frac{1}{x+b} = \frac{1}{c}$
 $\Rightarrow \frac{(x+b) + (x+a)}{(x+a)(x+b)} = \frac{1}{c}$
 $\Rightarrow 2cx + (a+b)c = x^2 + (a+b)x + ab$
 $\Rightarrow x^2 + (a+b-2c)x + ab - ac - bc = 0$
Let the roots of above equation be α and β .
Given, $\alpha + \beta = 0$
 $\Rightarrow -(a+b-2c) = 0 \Rightarrow a+b = 2c \dots (i)$
Now, $\alpha\beta = ab - ac - bc$
 $= ab - (a+b)c$
 $= ab - (a+b) \frac{(a+b)}{2}$
[from Eq. (i)]
 $= \frac{2ab - (a^2 + b^2 + 2ab)}{2} = -\frac{(a^2 + b^2)}{2}$

55. (b) Let the roots of the equation $kx^2 - 5x + 6 = 0$ be α and β .
 $\therefore \alpha + \beta = \frac{5}{k}$ and $\alpha\beta = \frac{6}{k}$
Given, $\frac{\alpha}{\beta} = \frac{2}{3} \Rightarrow \alpha = \frac{2}{3}\beta$
 $\therefore \frac{2}{3}\beta + \beta = \frac{5}{k}$ and $\frac{2}{3}\beta^2 = \frac{6}{k}$
 $\Rightarrow \frac{5}{3}\beta = \frac{5}{k}$ and $\beta^2 = \frac{9}{k}$
 $\beta = \frac{3}{k}$ and $\beta^2 = \frac{9}{k} \Rightarrow \frac{9}{k^2} = \frac{9}{k}$
 $\Rightarrow k = 1$ and $k \neq 0$
[$\therefore$ leading coefficient is never zero.]

56. (d) Let α and β be the roots of the equation $7x^2 + 12x + 18 = 0$.
 $\therefore \alpha + \beta = \frac{-12}{7}$ and $\alpha\beta = \frac{18}{7}$
 $\Rightarrow \alpha^2 + \beta^2 + 2\alpha\beta = \frac{144}{49}$
 $\Rightarrow \alpha^2 + \beta^2 = \frac{144}{49} - \frac{36}{7} = -\frac{108}{49}$
 $\therefore \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{-\frac{108}{49}}{\frac{18}{7}} = -\frac{6}{7} = -6 : 7$

57. (d) Given, $\frac{x(x-1) - (m+1)}{(x-1)(m-1)} = \frac{x}{m}$
 $\Rightarrow m(x^2 - x - m + 1) = x(mx - x - m + 1)$
 $\Rightarrow mx^2 - mx - m(m+1) = mx^2 - x^2 - mx + x$
 $\Rightarrow x^2 - x - m(m+1) = 0$
Let one root be α , then other root is also α .
 $\therefore \alpha + \alpha = 1, \Rightarrow \alpha = \frac{1}{2}$

$$\begin{aligned}\text{Also, } \alpha^2 &= -m(m+1) \\ \Rightarrow \left(\frac{1}{2}\right)^2 &= -m(m+1) \\ \Rightarrow 4m^2 + 4m + 1 &= 0 \\ \Rightarrow (2m+1)^2 &= 0 \\ \therefore m &= -\frac{1}{2}\end{aligned}$$

58. (c) Given, $3^x + 27(3^{-x}) = 12$
Let $\frac{3^x}{27} = y$
 $\therefore y + \frac{1}{y} = 12$
 $\Rightarrow y^2 - 12y + 27 = 0$
 $\Rightarrow y^2 - 9y - 3y + 27 = 0$
 $\Rightarrow (y-3)(y-9) = 0 \Rightarrow y = 3, 9$
 $\Rightarrow 3^x = 3$ or $3^x = 9 = 3^2$
 $\therefore x = 1$ or $x = 2$

59. (a) Let the roots of the given equation be α and β .
 $\therefore \alpha + \beta = a$ and $\alpha\beta = b$
Now, $|\alpha - \beta| = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta}$
 $= \sqrt{a^2 - 4b}$

60. (a) $x^2 - x - 2 = 0$
 $(x-2)(x+1) = 0 \Rightarrow x = 2, -1$
So, both the roots are integers.

61. (c) $x^2 - 6x + 5 = 0 \Rightarrow (x-5)(x-1) = 0$
 $\Rightarrow x = 5$ or 1
Also, $|x-3| = 2 \Leftrightarrow (x-3) = 2$
or $-(x-3) = 2 \Rightarrow x = 5$ or $x = 1$
 $\therefore x^2 - 6x + 5 = 0$
and $|x-3| = 2$ are equivalent.

62. (a) Let the number are x and $\frac{1}{x}$.
Then, $x + \frac{1}{x} = \frac{10}{3}$, $\frac{x^2 + 1}{x} = \frac{10}{3}$
 $\Rightarrow 3x^2 - 10x + 3 = 0$
 $\Rightarrow 3x^2 - 9x - x + 3 = 0$
 $\Rightarrow 3x(x-3) - 1(x-3) = 0$
 $\Rightarrow (3x-1)(x-3) = 0$
 $\therefore x = \frac{1}{3}, x = 3$

63. (b) Let the smaller part = x and greater part = $16 - x$
By given condition,
 $2(16-x)^2 - x^2 = 164$
 $\Rightarrow 2(256 + x^2 - 32x) - x^2 = 164$
 $\Rightarrow x^2 - 64x + 348 = 0$
 $\Rightarrow (x-58)(x-6) = 0$
 $\Rightarrow x = 58, x = 6$
Here $x \neq 58$
 $\therefore x = 6$
and, hence larger part
 $= 16 - x = 16 - 6 = 10$

64. (c) Here, $y = 15$, then $15 = \frac{x(x-1)}{2}$
 $\Rightarrow x^2 - x - 30 = 0$
 $\Rightarrow x^2 - 6x + 5x - 30 = 0$
 $\Rightarrow x(x-6) + 5(x-6) = 0$
 $\Rightarrow (x+5)(x-6) = 0$

Either $x = 6$ or $x = -5$
But $x \neq -5$, so $x = 6$
Thus, figure has 6 points.

65. (a) Let the natural numbers be x and $x+1$, respectively.
Then, $x^2 + (x+1)^2 = 221$
 $\Rightarrow 2x^2 + 2x + 1 = 221$
 $\Rightarrow 2x^2 + 2x - 220 = 0$
 $\Rightarrow x^2 + x - 110 = 0$
 $\Rightarrow (x+11)(x-10) = 0$
 $\Rightarrow x = -11$ and $x = 10$
 $x \neq -11$ [$\because$ numbers are natural]
So, for $x = 10$, next consecutive natural number
 $= x + 1 = 10 + 1 = 11$

66. (b) Let the consecutive positive odd integers be $2x+1$ and $2x+3$,
so $(2x+1)^2 + (2x+3)^2 = 130$
 $\Rightarrow (4x^2 + 4x + 1) + (4x^2 + 12x + 9) = 130$
 $\Rightarrow x^2 + 2x - 15 = 0$
 $\Rightarrow (x+5)(x-3) = 0$
 $\Rightarrow x = 3, x = -5$, but $x \neq -5$
[$\because$ integers are positive.]

$\therefore$ Two consecutive integers are
 $2x+1 = 7$ and $2x+3 = 9$
i.e. 7 and 9.

67. (b) Here, $2x+1 \geq 7 \Rightarrow 2x \geq 7-1$
 $\Rightarrow 2x \geq 6 \Rightarrow x \geq 3$

68. (c) Here, $\frac{x-1}{3} \geq 4$
 $\Rightarrow x-1 \geq 12 \Rightarrow x \geq 13$

69. (b) Here, $3x+2 \leq 5x-(4-x)$
 $\Rightarrow 3x+2 \leq 5x-4+x$
 $\Rightarrow 3x+2 \leq 6x-4$
 $\Rightarrow 3x-6x \leq -4-2 \Rightarrow -3x \leq -6$
 $\Rightarrow -x \leq -2 \Rightarrow x \geq 2$

70. (c) Here, $2x+3 \geq 8 \Rightarrow 2x \geq 8-3$
 $\Rightarrow 2x \geq 5 \Rightarrow x \geq \frac{5}{2}$

Again, $3x+1 \leq 12$

$\Rightarrow 3x \leq 11 \Rightarrow x \leq \frac{11}{3}$

By combining values, we get
 $\frac{5}{2} \leq x \leq \frac{11}{3}$

71. (b) $\frac{1}{2} \left(\frac{3}{5}x + 4 \right) \geq \frac{1}{3}(x-6)$
 $\Rightarrow \frac{3}{10}x + 2 \geq \frac{1}{3}x - 2$
 $\Rightarrow 9x + 60 \geq 10x - 60$
 $\Rightarrow -x \geq -120$
[multiplying both sides by -1]
 $\Rightarrow x \leq 120$

Thus, all real numbers x which are less than or equal to 120 satisfies the inequality.

72. (a) We have, $4x^2 - 1 \leq 0$
 $\Rightarrow (2x)^2 - 1 \leq 0$
 $\Rightarrow (2x-1)(2x+1) \leq 0$
So, either $(2x-1) \geq 0$
and $(2x+1) \leq 0$... (i)
or $(2x-1) \leq 0$
and $(2x+1) \geq 0$... (ii)
From Eq. (i), $2x \geq 1$ and $2x \leq -1$
 $\Rightarrow x \geq \frac{1}{2}$ and $x \leq \frac{-1}{2}$

which is not possible.

From Eq. (ii), $(2x-1) \leq 0$
and $(2x+1) \geq 0$

$2x \leq 1$ and $2x \geq -1$
 $x \leq \frac{1}{2}$ and $x \geq \frac{-1}{2}$
 $\therefore \frac{-1}{2} \leq x \leq \frac{1}{2}$ is the required solution set.

73. (b) Here, $3x+2y = 12$ passes through the points $(0, 6)$ and $(4, 0)$. So, its graph is line AB . Also, $3x+2y \leq 12$ as the shaded region is below AB .
 $x = 0$ and $y = 0$ are the Y -axis and X -axis, respectively. $x \geq 0$ and $y \geq 0$ implies the region on the right hand side of Y -axis and region above X -axis, respectively.

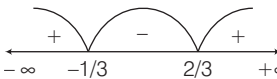
The three portion $x \geq 0, y \geq 0$ and $3x+2y \leq 12$ intersects to give the shaded portion OAB .

74. (d) Given, $x^2 + 3x - 10 > 0$
 $\Rightarrow x^2 + 5x - 2x - 10 > 0$
 $\Rightarrow (x+5)(x-2) > 0$
Either, $(x+5) > 0$ and $(x-2) > 0$
or $(x+5) < 0$ and $(x-2) < 0$
 $\Rightarrow x > 2$ or $x < -5$

75. (a) $x^2 - 5x + 6 \geq 0$
 $\Rightarrow (x-2)(x-3) \geq 0$

 $\Rightarrow x \leq 2$ or $x \geq 3$
 $\Rightarrow x \in [-\infty, 2] \text{ or } [3, \infty]$
 $\Rightarrow x \in [-\infty, 2] \cup [3, \infty]$

76. (b) $\frac{x-2}{3x+1} < \frac{x-3}{3x-2}$
 $\Leftrightarrow \frac{x-2}{3x+1} - \frac{x-3}{3x-2} < 0$
 $\Leftrightarrow \frac{(x-2)(3x-2) - (x-3)(3x+1)}{(3x+1)(3x-2)} < 0$
 $\Leftrightarrow \frac{7}{(3x+1)(3x-2)} < 0$
 $\Leftrightarrow (3x+1)(3x-2) < 0$
 $\Leftrightarrow 3\left(x + \frac{1}{3}\right)3\left(x - \frac{2}{3}\right) < 0$
 $\Leftrightarrow \left(x + \frac{1}{3}\right)\left(x - \frac{2}{3}\right) < 0$



$\Leftrightarrow -\frac{1}{3} < x < \frac{2}{3}$
 So, $x \in \left[-\frac{1}{3}, \frac{2}{3}\right]$

77. (c) Here, as $x > 0$ and $y > 0$ so both are positive and satisfies $x + 2y \leq 3$.
 $\therefore$ When, $x = 3$ and $y = 1$, we get $x + 2y = 5$.

Clearly, this does not satisfy $x + 2y \leq 3$

When $x = 1$ and $y = 1$,

then, $x + 2y \leq 3$.

So, $x = 1, y = 1$ is one of the solutions.

78. (d) As, $(0, 0)$ satisfies $x + 3y \leq 3$.

And $(0, 0)$ satisfies $x + y \leq 2$.

Clearly, shaded region is the position common to the line $x + 3y = 3$ and below it and that on the line $x + y = 2$ and below it.

So, shaded region is the solution set

$$x + y \leq 2, x + 3y \leq 3, x \geq 0, y \geq 0.$$

79. (c) Given, $a + b = 2m^2$... (i)

$$b + c = 6m \quad \dots (ii)$$

$$\text{and } a + c = 2 \quad \dots (iii)$$

On adding Eqs. (i), (ii) and (iii), we get

$$2(a + b + c) = 2m^2 + 6m + 2$$

$$\Rightarrow a + b + c = m^2 + 3m + 1 \quad \dots (iv)$$

On subtracting Eq. (ii) from Eq. (iv), we get

$$a = m^2 - 3m + 1$$

On subtracting Eq. (iii) from Eq. (iv), we get

$$b = m^2 + 3m - 1$$

On subtracting Eq. (i) from Eq. (iv), we get

$$c = -m^2 + 3m + 1$$

As, $a \leq b$ and $b \leq c$

$$\Rightarrow m^2 - 3m + 1 \leq m^2 + 3m - 1$$

$$\text{and } m^2 + 3m - 1 \leq -m^2 + 3m + 1$$

$$\Rightarrow 6m \geq 2 \text{ and } 2m^2 \leq 2$$

$$\Rightarrow m \geq \frac{1}{3} \text{ and } -1 \leq m \leq 1$$

$$\therefore \frac{1}{3} \leq m \leq 1$$

80. (d) Given equation, $x^2 - 3x + 2 = 0$

$$\Rightarrow x^2 - 2x - x + 2 = 0$$

$$\Rightarrow (x-2)(x-1) = 0 \Rightarrow x = 2, 1$$

Let $\alpha = 1$ and $\beta = 2$

$$\therefore \alpha + 1 = 2 \text{ and } (\beta + 1) = 3.$$

Now, sum of roots = $2 + 3 = 5$

and product of roots = $2 \times 3 = 6$

$\therefore$ Required equation

$$= x^2 - (\text{sum of roots}) + \text{product of roots} = 0$$

$$\Rightarrow x^2 - 5x + 6 = 0$$

Hence, the equation is neither I nor II.

81. (c) I. Every quadratic equation has two roots, which may or may not be real.

II. $x^2 - 4x + 2 = 0$ has integral coefficients but does not have integral roots.

III. Since, discriminant = $b^2 - 4ac$

If a and c have opposite sign, then $b^2 - 4ac \geq 0$

$\therefore$ The quadratic equation has real roots

82. (c) I. $x^2 + \frac{1}{x^2} = 2$

or $x^4 - 2x^2 + 1 = 0$
 is not a quadratic equation.

II. $x + \frac{3}{x} = x^2$

$$\Rightarrow \frac{x^2 + 3}{x} = x^2$$

$$\text{or } x^2 + 3 = x^3$$

$$\Rightarrow x^3 - x^2 - 3 = 0$$

is not a quadratic equation.

III. $2x^2 - x + 2 = x^2 + 4x - 4$

$$\text{or } 2x^2 - x + 2 - x^2 - 4x + 4 = 0$$

$$\text{or } x^2 - 5x + 6 = 0$$

is a quadratic equation.

IV. $x^3 + 6x^2 + 2x - 1 = 0$

is not a quadratic equation.

Hence, only III statement is correct.

83. (b) I. $3x^2 - 2x - 1 = 0$

$$\text{Put } x = 1 \Rightarrow 3(1)^2 - 2(1) - 1 = 0$$

$$\therefore \text{LHS} = \text{RHS}$$

II. $x^2 + \sqrt{2}x - 4 = 0$

$$\text{Put } x = -2\sqrt{2}$$

$$\Rightarrow (-2\sqrt{2})^2 + \sqrt{2}(-2\sqrt{2}) - 4 = 0$$

$$\therefore \text{LHS} = \text{RHS}$$

III. $x^2 + x + 1 = 0; x = 1, x = -1$

Put $x = 1$, we get $1^2 + 1 + 1 = 3 \neq 0$

$\therefore$ LHS $\neq$ RHS

$\therefore x = 1$ is not a solution.

Put $x = -1$, we get

$$(-1)^2 + (-1) + 1 = 1 - 1 + 1 = 1 \neq 0$$

$\therefore$ LHS $\neq$ RHS

$\therefore x = -1$ is not a solution.

IV. $9x^2 - 3x - 2 = 0$

Put $x = -1/3$

$$\Rightarrow 9(-1/3)^2 - 3(-1/3) - 2 = 0$$

$\therefore$ LHS = RHS

Now, put $x = 2/3$

$$\Rightarrow 9(2/3)^2 - 3(2/3) - 2 = 0$$

$\therefore$ LHS = RHS

84. (b) Given, $h = -16t^2 + 64t + 80$

put $t = 1$

$$\Rightarrow h = -16(1)^2 + 64(1) + 80 = 128 \text{ m}$$

85. (c) By rearranging, we get

$$h = -16(t^2 - 4t - 5)$$

$$\Rightarrow h = -16[(t-2)^2 - 9]$$

$$\Rightarrow h = -16(t-2)^2 + 144$$

when the height is maximum, $t = 2$

$\therefore$ maximum height = 144 m

86. (a) When the ball hits the ground $h = 0$

$$\Rightarrow -16t^2 + 64t + 80 = 0$$

$$\Rightarrow t^2 - 4t - 5 = 0$$

$$\Rightarrow t^2 - 5t + t - 5 = 0$$

$$\Rightarrow (t-5)(t+1) = 0 \Rightarrow t = 5, -1$$

Since, the time cannot be negative.

So, $t = 5$ sec

87. (a) Let $f(x) = x^2 - 2ax + a^2 + a - 3$

$$f(3) > 0 \Rightarrow 9 - 6a + a^2 + a - 3 > 0 \dots (i)$$

Also, since roots are real

$$\therefore B^2 - 4AC \geq 0$$

$$\Rightarrow 4a^2 - 4(a^2 + a - 3) \geq 0 \dots (ii)$$

From Eqs. (i) and (ii), we have

$$a^2 - 5a + 6 > 0 \text{ and } a - 3 \leq 0$$

$$\Rightarrow a^2 - 3a - 2a + 6 > 0$$

$$\Rightarrow (a-3)(a-2) > 0$$

$$\Rightarrow a \leq 3 \text{ and } a < 2 \text{ or } a > 3$$

Combining the above inequalities, we have $a < 2$

88. (a) Given, $7x^2 - 50x + k = 0$

Here, $a = 7, b = -50$ and $c = k$

Since, α and β are the roots of the given equation

$$\therefore \alpha + \beta = \frac{-b}{a} \Rightarrow \alpha + \beta = \frac{50}{7}$$

$$\Rightarrow \beta = \frac{1}{7} \quad [\because \alpha = 7, \text{ given}]$$

$$\text{and } \alpha\beta = \frac{c}{a} \Rightarrow 7 \times \frac{1}{7} = \frac{k}{7} \Rightarrow k = 7$$

89. (b) Let α and β be the roots of the quadratic equation

$$x^2 + px + q = 0.$$

Given that, A starts with a wrong value of p and obtains the roots as 2 and 6. But this time q is correct.

i.e. Product of roots

$$= q = \alpha \cdot \beta = 6 \times 2 = 12 \quad \dots(i)$$

and B starts with a wrong value of q and gets the roots as 2 and -9. But this time p is correct.

i.e. Sum of roots

$$= p = \alpha + \beta = -9 + 2 = -7 \quad \dots(ii)$$

We know that,

$$\begin{aligned} (\alpha - \beta)^2 &= (\alpha + \beta)^2 - 4\alpha\beta \\ &= (-7)^2 - 4 \times 12 = 49 - 48 = 1 \end{aligned}$$

[from Eqs. (i) and (ii)]

$$\Rightarrow \alpha - \beta = 1 \quad \dots(iii)$$

From Eqs. (ii) and (iii), $\alpha = -3$ and $\beta = -4$

which are correct roots.

90. (a) Let one root be α , then other roots is α^2

Given equation is $x^2 - bx + c = 0$

$$\therefore \text{Sum of roots} = \alpha + \alpha^2 = -\frac{(-b)}{1}$$

$$\Rightarrow \alpha(\alpha + 1) = b \quad \dots(i)$$

$$\text{and product of roots} = \alpha \cdot \alpha^2 = \frac{c}{1}$$

$$\Rightarrow \alpha^3 = c \Rightarrow \alpha = c^{1/3} \quad \dots(ii)$$

From Eqs. (i) and (ii),

$$c^{1/3} (c^{1/3} + 1) = b \quad \dots(iii)$$

On cubing both sides, we get

$$c(c^{1/3} + 1)^3 = b^3$$

$$\Rightarrow c\{c + 1 + 3c^{1/3}(c^{1/3} + 1)\} = b^3$$

$$\Rightarrow c(c + 1 + 3b) = b^3 \quad [\text{from Eq. (iii)}]$$

$$\Rightarrow b^3 = 3bc + c^2 + c$$

91. (a) Let α and β be the roots of the given quadratic equation $2x^2 - 11x + 5 = 0$.

$$\therefore \alpha + \beta = -\frac{(-11)}{2} = \frac{11}{2} \quad \dots(i)$$

$$\text{and } \alpha \cdot \beta = \frac{5}{2} \quad \dots(ii)$$

Now, $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$$\begin{aligned} &= \left(\frac{11}{2}\right)^2 - 4\left(\frac{5}{2}\right) \\ &= \frac{121}{4} - \frac{40}{4} = \frac{81}{4} = \left(\frac{9}{2}\right)^2 \end{aligned}$$

$$\therefore \text{Difference of roots} = (\alpha - \beta) = \frac{9}{2} = 4.5$$

92. (b) Let the two numbers be x and y .

Now, by given condition,

$$x^2 + y^2 = 97 \quad \dots(i)$$

$$\text{and } (x - y)^2 = 25 \quad \dots(ii)$$

$$\Rightarrow (x^2 + y^2) - 2xy = 25$$

$$\Rightarrow 97 - 2xy = 25 \quad [\text{from Eq. (i)}]$$

$$\Rightarrow 2xy = 72$$

$$\therefore xy = 36$$

$$93. (a) \text{ Given that, } x + \frac{1}{x} = 2 \quad \dots(i)$$

On squaring both sides, we get

$$\left(x + \frac{1}{x}\right)^2 = 4 \Rightarrow x^2 + \frac{1}{x^2} + 2 = 4$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 2 \quad \dots(ii)$$

$$\begin{aligned} \text{Now, } \left(x - \frac{1}{x}\right)^2 &= \left(x^2 + \frac{1}{x^2}\right) - 2 \\ &= 2 - 2 = 0 \quad [\text{from Eq. (ii)}] \end{aligned}$$

$$\therefore x - \frac{1}{x} = 0$$

94. (b) Let the number of columns be x .

Then, number of rows = $x + 4$

According to the question,

$$x(x + 4) \times 5 - 2 = 158$$

$$\Rightarrow 5x(x + 4) = 160 \Rightarrow x(x + 4) = 32$$

$$\Rightarrow x^2 + 4x - 32 = 0$$

$$\Rightarrow x^2 + 8x - 4x - 32 = 0$$

$$\Rightarrow x(x + 8) - 4(x + 8) = 0$$

$$\Rightarrow (x + 8)(x - 4) = 0$$

So, $x = 4$ as $x = -8$ is not possible.

$\therefore$ Number of rows

$$= x + 4 = 4 + 4 = 8$$

$$95. (c) \text{ Given, } x^2 + \frac{1}{x^2} + 8\left(x + \frac{1}{x}\right) + 14$$

$$\text{Let } x + \frac{1}{x} = y \quad \dots(i)$$

On squaring both sides, we get

$$\therefore \left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 = y^2$$

$$\Rightarrow x^2 + \frac{1}{x^2} = y^2 - 2 \quad \dots(ii)$$

Now, on putting values from Eqs. (i) and

(ii) in the given equation, we get

$$y^2 - 2 + 8y + 14 = y^2 + 8y + 12$$

$$= y^2 + 2y + 6y + 12$$

$$= y(y + 2) + 6(y + 2)$$

$$= (y + 6)(y + 2)$$

$$= \left(x + \frac{1}{x} + 6\right)\left(x + \frac{1}{x} + 2\right)$$

$$\text{So, one factor is } \left(x + \frac{1}{x} + 6\right).$$

96. (b) Given, $x^2 - x - 1 = 0$

Here, $a = 1$, $b = -1$ and $c = -1$

$\therefore \alpha$ and β are the roots of the equation

$$x^2 - x - 1 = 0$$

$$\therefore \alpha + \beta = \frac{-b}{a} = 1 \quad \text{and} \quad \alpha\beta = \frac{c}{a} = -1$$

$$\frac{\alpha^2 + \beta^2}{(\alpha^2 - \beta^2)(\alpha - \beta)} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha - \beta)^2(\alpha + \beta)}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{[(\alpha + \beta)^2 - 4\alpha\beta](\alpha + \beta)}$$

$$= \frac{(1)^2 - 2 \times -1}{[(1)^2 - 4 \times -1] \times 1} = \frac{3}{5}$$

97. (d) Given,

$$x^2 = 6 + \sqrt{6 + \sqrt{6 + \sqrt{6 + \dots \infty}}}$$

$$\text{So, } x^2 = 6 + \sqrt{x^2} \Rightarrow x^2 = 6 + x$$

$$\Rightarrow x^2 - x - 6 = 0$$

$$\Rightarrow x^2 + 2x - 3x - 6 = 0$$

$$\Rightarrow x(x + 2) - 3(x + 2) = 0$$

$$\Rightarrow (x - 3)(x + 2) = 0$$

$$\therefore x = 3, -2$$

98. (d) Given, $ax^2 + bx + b = 0$

$$\Rightarrow x^2 + \frac{b}{a}x + \frac{b}{a} = 0$$

$$\therefore \text{Sum of roots, } \alpha + \beta = \frac{-b}{a}$$

$$\text{and products of roots, } \alpha\beta = \frac{b}{a}$$

Hence, product of roots is not equal to the sum of roots,

So, statement I is not correct.

Now, for roots to be real and unequal, $D > 0$

$$\Rightarrow b^2 - 4ac > 0$$

$$\Rightarrow b^2 - 4a(b) > 0 \quad [\because c = b]$$

$$\Rightarrow b^2 - 4ab > 0 \Rightarrow b^2 > 4ab$$

$$\therefore b > 4a$$

So, if $b > 4a$, then roots are unequal and real, so statement II is not always true, it depends on values of a and b .

99. (d) Given, $Ax^2 + Bx + C = 0 \quad \dots(i)$

Since, the given roots are -1 and 1.

$$\therefore \text{Sum of roots} = -1 + 1 = 0$$

$$\text{and product of roots} = 1 \times (-1) = -1$$

$$\Rightarrow \text{quadratic equation is } x^2 -$$

$$(\text{sum of roots})x + \text{product of roots} = 0$$

$$\Rightarrow x^2 - 0 \cdot x - 1 = 0 \Rightarrow x^2 - 1 = 0$$

On comparing with above equation from Eq. (i), we get

$$A = 1 \text{ and } C = -1$$

So, A and C are of opposite sign.

100. (d) Given, $(a^2 - bc)x^2 + 2(b^2 - ac)x + (c^2 - ab) = 0$

Since, the given roots are equal.

$$\therefore D = 0$$

$$\text{i.e. } [2(b^2 - ac)]^2 - 4(a^2 - bc)$$

$$(c^2 - ab) = 0$$